

Sprint 2

Accessing SAP HANA Cloud and Administering SAP HANA Cloud

Structured study notes with definitions, key points, demos, exercises, SQL/CLI snippets, and original visual diagrams

Which Administration Tool Should I Use?

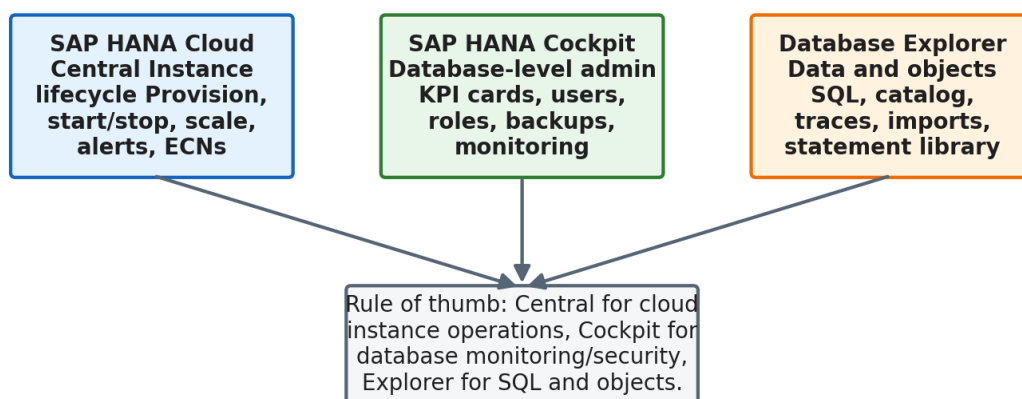


Figure 1. Original study diagram: administration tool selection map.

How to use this pack: Start with the quick reference and glossary, then work through each module. For each module, read the objective, definitions, key points, process steps, and complete the listed demo or exercise. Finish with the self-check questions and command cheat sheets.

Image note: The diagrams in this document are original study visuals created from the supplied lesson text. They are not SAP product screenshots.

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Quick Reference: What to Remember First

Topic	Must-know summary
Account choice	Enterprise edition is for business/production use; trial is for individual learning; free tier plans are for customer/partner/organizational POCs inside an enterprise account.
Main admin hub	SAP HANA Cloud Central is the preferred hub for SAP HANA Cloud instance administration and monitoring.
Tool split	Cloud Central = instance operations; HANA cockpit = database-level monitoring/admin/security; database explorer = SQL, catalog, objects, diagnostic files, imports.
Free tier/trial limits	A typical free tier/trial HANA database is limited to one instance with 16 GB RAM, 1 vCPU, and 80 GB storage; data lake is 32 GB RAM, 4 vCPUs, and 256 GB storage; services are stopped daily around midnight.
Alerting	Alerts are generated when configurable thresholds are exceeded. Alert rules define checks; alert rule sets group rules for reuse; SAP BTP Alert Notification Service handles external notifications.
Backups	SAP performs automatic backups. The default cycle includes daily complete data backups, log/catalog backups every 15 minutes, and usually a 14-day retention period.
Recovery	Point-in-time recovery should target a time before the incident. Investigate root cause before recovery.
ECN	Elastic Compute Nodes add temporary compute capacity for peak workloads and can be used with workload classes, routing hints, and optional table replication.
Security	Use roles rather than direct privilege grants where possible. User groups organize user administration but do not control data access.
Joule	Joule with limited features can navigate apps, summarize alerts, generate SQL, and change settings, but output must be verified before use.

Glossary of Essential Definitions

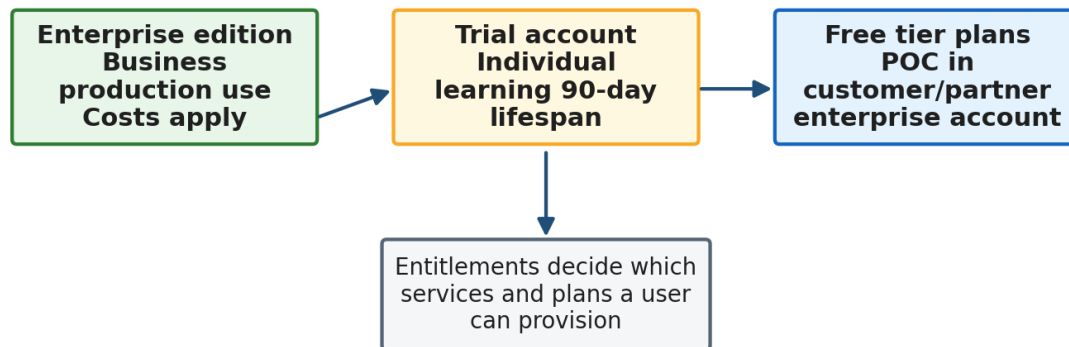
Term	Definition
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SAP BTP	SAP Business Technology Platform; the cloud platform layer that hosts accounts, subaccounts, services, runtimes, and SAP HANA Cloud.
Global account	Top-level SAP BTP account that contains subaccounts and entitlements.
Subaccount	A unit inside a global account, normally mapped to one region, where services are subscribed to and provisioned.
Entitlement	Permission/allowance assigned to a subaccount or user to use a specific SAP BTP service or plan.
Service plan	A particular commercial or technical variant of a service, such as hana, hana-free, tools, or relational-data-lake-free.
Cloud Foundry runtime	Application runtime environment used in SAP BTP and accessed through the Cloud Foundry CLI.
SAP HANA Cloud Central	Central application for administering and monitoring SAP HANA Cloud database and data lake instances.
SAP HANA cockpit	Database administration and monitoring tool for resources, alerts, backups, users, roles, and configuration.
SAP HANA database explorer	Tool for SQL execution, catalog object browsing, diagnostic files, imports/exports, and statement library usage.
Integrated data lake	SAP HANA Cloud, data lake attached to an existing SAP HANA Cloud database instance.
Standalone data lake	Independent SAP HANA Cloud, data lake instance provisioned separately in SAP BTP.
Availability zone	Separate failure domain inside a region, with independent power, network, and cooling.
Replica	Copy of a SAP HANA Cloud database used to increase availability and/or disaster tolerance.
RPO	Recovery Point Objective; maximum acceptable amount of data loss measured in time.
Alert rule	A monitoring check with threshold values, status, and check interval.
Alert rule set	A reusable group of alert rules applied to one or more database instances.
Statement Library	Store of reusable SQL statements, including user-defined SQL and SAP-delivered statements.
Trace file	Diagnostic log/trace file used to analyze database errors and internal issues.
Elastic Compute Node	Temporary additional SAP HANA Cloud compute capacity used to handle workload peaks.
Workload class	SAP HANA mechanism for classifying and controlling workloads, including routing to ECNs.
Audit policy	Rule defining which database actions should be written to the audit trail.
Audit trail	Persistent record of audited database events.
Standard user	Database user created with CREATE USER; can create objects in own schema and has PUBLIC role by default.
Restricted user	Database user created with CREATE RESTRICTED USER; initially has no privileges and is intended for application access.
Role	Reusable collection of privileges assigned to users or other roles.
System privilege	Privilege that controls general system activities such as creating users, roles, schemas, or backups.
Object privilege	Privilege that controls actions on database objects such as tables, views, and schemas.
Analytic privilege	Privilege controlling read access to SAP HANA information model data based on values.
Joule	SAP AI assistant interface used in SAP HANA Cloud Central with limited features for operational assistance.

Module 1 - SAP BTP Account Types and Access

Objective: Distinguish SAP BTP account types and select the correct access model for learning, proof of concept, or business use.

SAP BTP Account Choice



Use the account type that matches the business context, budget, and learning objective.

Figure 2. Account types and when to use them.

Account type	Best used for	Important limitations / notes
SAP BTP enterprise edition	Business users, administrators, implementation projects, production landscapes, and customer POCs.	Provisioning SAP HANA Cloud can incur cost. Free tier service plans can be used in customer/partner enterprise accounts for POCs.
SAP BTP trial account	Students and individual learners who need a guided, no-cost environment to learn SAP HANA Cloud.	Trial setup creates a global account, subaccount, and Cloud Foundry runtime automatically. Trial content is deleted after the trial period.
Free tier plans inside enterprise accounts	POCs where work should later be converted from free plan to paid hana plan without losing development content.	For organizations, customers, and partners; old individual free tier signup process is obsolete. Individual learners should use BTP Trial.

Key distinction: Trial is an account type for individual learning. Free tier is a service-plan model inside an enterprise account for organizational use and POCs.

Required entitlements: To use SAP HANA Cloud and the administration tools, the subaccount/user needs relevant entitlements such as hana, relational-data-lake, tools, and, for legacy database tools, MEMORY for the Cloud Foundry runtime.

- Use enterprise edition for real company landscapes and paid usage.
- Use trial when you do not have access to an enterprise account and need a learning environment.
- Use free tier service plans for a proof of concept in an enterprise account when you may later convert to a paid plan.
- Check region and entitlement availability before assuming that SAP HANA Cloud or data lake features can be provisioned.

Module 2 - Logging in and Required Entitlements

Objective: Sign in using SAP BTP Cockpit and understand access URLs, resource restrictions, and sign-in flow differences between enterprise and trial accounts.

Region / account	Cockpit URL
Enterprise - EMEA	https://emea.cockpit.btp.cloud.sap
Enterprise - Restricted EU Access	https://eu-access.cockpit.btp.cloud.sap
Enterprise - Americas	https://amer.cockpit.btp.cloud.sap
Enterprise - Asia-Pacific / Oceania	https://apac.cockpit.btp.cloud.sap
Trial	https://account.hanatrial.ondemand.com/trial/#/home/trial

Area	Free tier / trial resource pattern
SAP HANA database	One instance with 16 GB RAM, 1 vCPU, and 80 GB storage.
SAP HANA Cloud data lake	Can be connected with 32 GB RAM, 4 vCPUs, and 256 GB storage.
Daily stop behavior	HANA database and data lake are stopped daily around midnight.
Restart requirement	In free tier, you have 30 days to restart or the instance may be deleted. In trial, inactive accounts can be suspended and trial content is deleted at the end of the lifecycle.
Administration tools	SAP HANA cockpit and database explorer remain available for administration tasks attached to the instance.

Exam point: Enterprise accounts do not create SAP HANA Cloud trial databases. For test usage in enterprise accounts, add the free tier service plans instead.

1. Open the correct cockpit URL for your region or trial account.
2. Enter SAP ID / enterprise credentials.
3. Use two-factor authentication if configured.
4. Navigate to the global account and subaccount.
5. Verify entitlements before provisioning services.

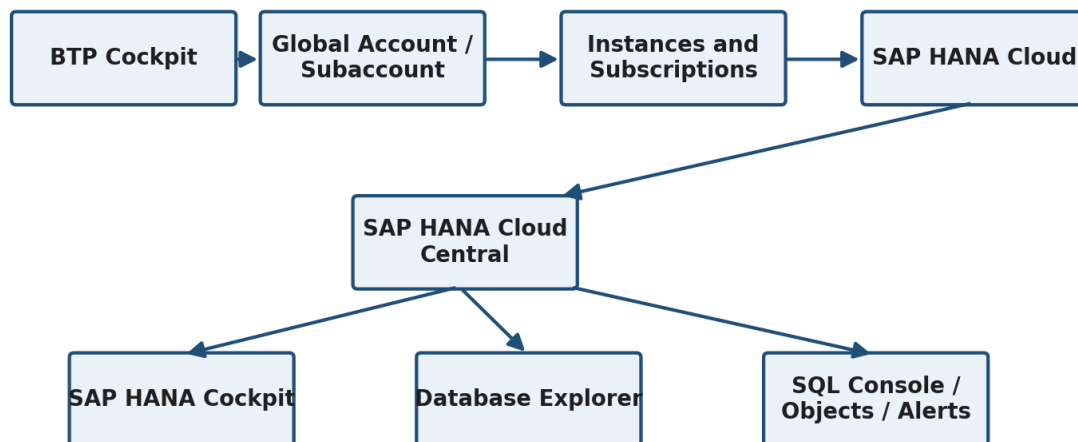
```
# Cloud Foundry login pattern
cf login -o <organisation> -s <space> -a <API endpoint>

# Trial accounts with one org and one space may not require -o, -s, or -a.
```

Module 3 - Administration Tools: Cloud Central, Cockpit, Database Explorer

Objective: Identify the three major SAP HANA Cloud administration tools and choose the correct one for each task.

Navigation to SAP HANA Cloud Administration



Cloud Central is the operational hub; Cockpit and Database Explorer open from instance actions.

Figure 3. BTP Cockpit navigation path to SAP HANA Cloud Central and related tools.

Which Administration Tool Should I Use?

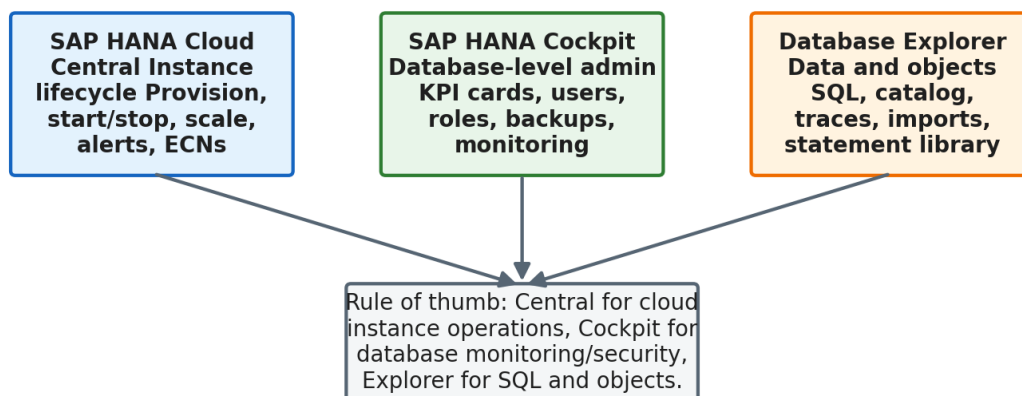


Figure 4. Tool selection map.

Tool	Primary purpose	Typical tasks
SAP HANA Cloud Central	Instance-level administration and monitoring across SAP HANA Cloud databases and data lakes.	Create instances, view status, start/stop/delete, scale, add data lake, check alerts, open cockpit/explorer, manage ECNs.
SAP HANA cockpit	Database-level administration and monitoring.	Monitor memory/CPU/storage, analyze SQL, manage users/roles, view backups, configure settings, inspect alerting cards.
SAP HANA database explorer	Direct work with database data, SQL, objects, and diagnostic files.	Run SQL, browse catalog, import/export catalog objects, view traces, open statement library, inspect HDI containers.

Activation step: Subscribe to the SAP HANA Cloud - Tools service in Instances and Subscriptions, assign the SAP HANA Cloud Administrator role collection, then log out and back in to refresh permissions.

- Cloud Central includes an integrated SAP HANA cockpit and launch points to database explorer.
- The database overview page contains KPI cards such as Memory, Compute, and Consumption.
- Cloud Central is currently the preferred administration location for SAP HANA Cloud database instances.
- Cockpit access remains available from Cloud Central but is noted as a capability that may be deprecated for some workflows over time.

Module 4 - Provisioning, Data Tiering, and Data Lake Options

Objective: Understand how SAP HANA Cloud database and data lake instances are provisioned, and how hot, warm, and cold storage options support cost-effective data management.

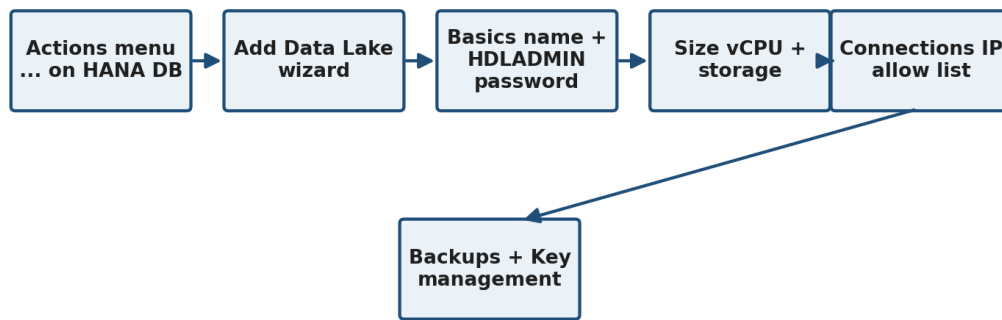
Provisioning choice	Use case
Preset configuration	Fast setup using SAP-recommended starting sizes.
JSON configuration file	Repeatable deployment of identical or standardized instances.
Configure manually	Full control over instance type, memory, storage, region/zone, replicas, backup, cloud connector, and key management.
Clone from existing template	Create a new database instance based on a cloning template from another instance.

Provisioning area	Key decision
Basics	Instance name, description, DBADMIN password, and HANA Cloud version/QRC.
Sizing	Performance class, memory, compute ratio, and storage.
Availability	Availability zone and optional replicas.
Connections	Allowed IP addresses and connection controls.
Backups and encryption	Backup schedule, key management, cloud connector, and customer-controlled encryption options.

Security checkpoint: DBADMIN is the initial super user for SAP HANA Cloud and should not be used for daily work. Create named users and roles for normal access.

Tier	Storage behavior	Best for	Cost/performance
Hot data	Stored in-memory.	Frequently changed data needed for real-time processing and analytics.	Highest performance and highest total cost.
Warm data	Managed on disk through Native Storage Extension (NSE), not fully loaded into memory.	Less frequently accessed operational data needing low latency.	More cost-effective than hot data.
Cold data	Managed separately, typically in integrated data lake, accessed through virtualization.	Rarely accessed, very large, long-term or analytic data.	Lowest cost option; supports massive scale.

Adding an Integrated Data Lake



Security checkpoint: Avoid "allow all IP addresses" unless explicitly required and approved.

Figure 5. Integrated data lake provisioning workflow.

Caution: Allowing all IP addresses exposes the data lake to the internet. Prefer SAP BTP-local connections or explicit company IP ranges unless there is a controlled business requirement.

Data lake type	Description	When to use
Integrated data lake	Attached to an existing SAP HANA Cloud database instance.	When HANA and data lake should work together and cost efficiency matters.
Standalone data lake	Independent data lake instance in SAP BTP.	When BI or analytics teams need a separate large-scale store, including unstructured data use cases.

Module 5 - Monitoring, Alerts, SQL Console, and Statement Library

Objective: Use Cloud Central monitoring capabilities to check alerts, configure alert behavior, monitor performance, and run or store SQL statements.

Alert Monitoring and Notification Flow

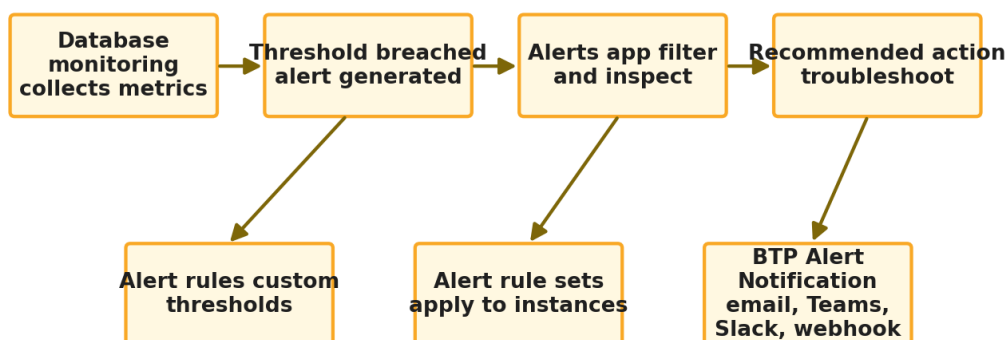


Figure 6. Alert monitoring, alert rules, alert rule sets, and notification flow.

Alert severity	Meaning	Typical response
Error	Immediate action required to mitigate downtime, data loss, data corruption, or system failure.	Investigate immediately. Use details, recommended action, resource usage, and traces.
Warning	Short-term action required, typically within hours or days, to reduce risk of future downtime.	Plan near-term correction and monitor trend.
Notice	Medium-term action required to prevent future potential issues.	Add to maintenance or capacity planning.
Information	Recommended action for performance or stability improvement.	Review during regular health checks.

- The Alerts app defaults to showing Error and Warning alerts from the past 24 hours.
- Selecting an alert displays a description and recommended action.
- Alert rules have default thresholds and check intervals that can be edited to align with company standards.
- Alert rule sets group multiple alert rules and can be saved or saved-and-applied to selected instances.
- Email and external notifications are handled through SAP BTP Alert Notification Service, not directly in Cloud Central Alerts.

Notification targets: SAP BTP Alert Notification Service can send e-mail, Microsoft Teams or Slack messages, ServiceNow tickets, or webhook calls to REST endpoints.

SQL Console and Statement Library Workflow

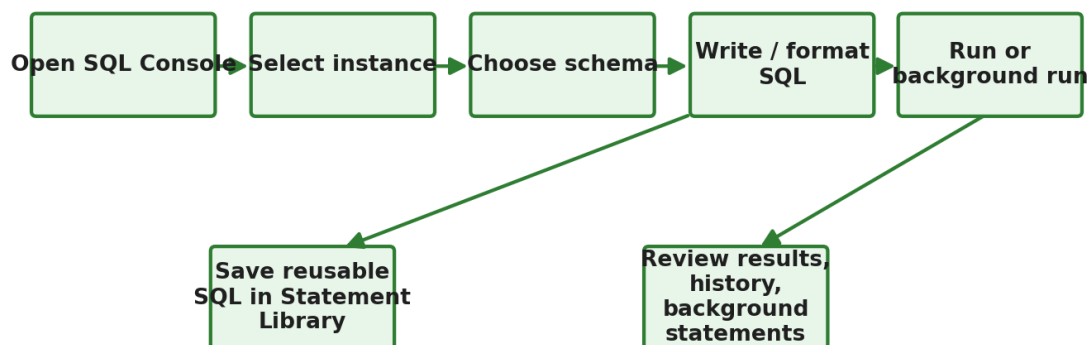


Figure 7. SQL Console and Statement Library workflow.

SQL Console area	Purpose
Select an Instance	Connects the SQL Console to a database instance you are authorized to access.
Stored credentials	Uses database user credentials mapped to your SAP BTP user.
Different user	Connects with an explicit database username and password.
Schema selector	Defines the schema context for statement execution.
Run options	Run batch, run statement, run line, or run statement in background for long-running SQL.
Results / History / Background Statements	Displays output, execution metadata, and background execution status.

Statement Library: Save frequently used SQL with a useful name and description. User-saved statements are private to the user account; SAP system statements are provided for analysis and monitoring.

```
-- Example SQL generated or used during study
CREATE SCHEMA "HC200_SCHEMA";
SET SCHEMA "HC200_SCHEMA";

CREATE COLUMN TABLE "HC200_SCHEMA"."HC200_DEMO" (
  "ProductNumber" INT PRIMARY KEY,
  "ProductName" NVARCHAR(255),
  "Stock" INT
);

INSERT INTO "HC200_DEMO" VALUES (1, 'Bike', 15);

SELECT "ProductName", "Stock"
FROM "HC200_DEMO"
WHERE "Stock" < 5;

SELECT "ProductName", "Stock"
FROM "HC200_DEMO"
WHERE "ProductName" LIKE 'P%';
```

Module 6 - Database Explorer, Traces, Imports, and Cockpit Navigation

Objective: Open SAP HANA database explorer and SAP HANA cockpit, then use them for catalog objects, traces, imports, statement library access, and database overview monitoring.

Database Explorer area	Study notes
Navigation panel	Switch between assigned databases; add or remove database connections.
Catalog	Data dictionary containing adapters, indexes, libraries, procedures, data structures, tables, and schemas.
HDI Containers	Design-time and runtime container structures used by developers to store and execute development work.
Database Diagnostic Files	Log, trace, diagnosis, error, and information files used to analyze problems.
SQL Console	Execute SQL directly against the connected database.

Default database user: In SAP HANA database explorer, DBADMIN is used by default for database connections in the training scenario. In real environments, use named administrator accounts and least-privilege roles.

- To examine traces, drill down through Database Diagnostic Files to the service folder such as indexserver.
- Open the indexserver trace and search for errors or warnings with Ctrl+F.
- Right-click a diagnostic file to open, show, download, copy the file name, or create a shortcut.
- For SAP Support, generate a full system information dump ZIP from the cockpit database overview through Manage full system information dumps.

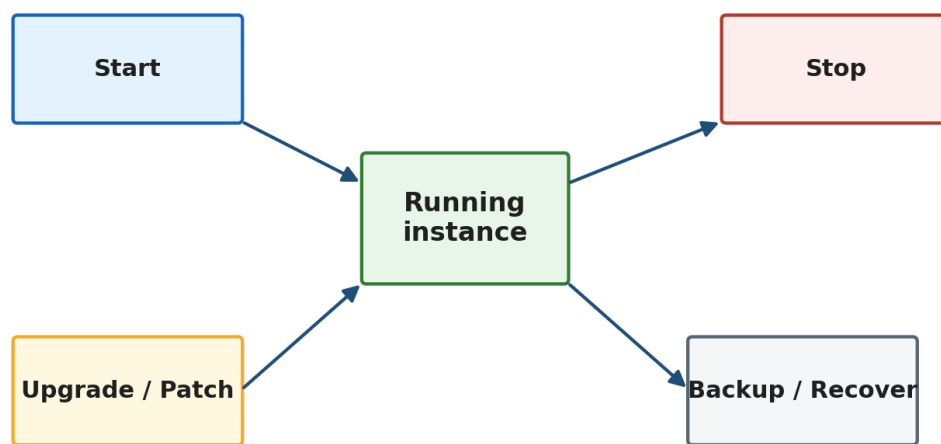
Import prerequisite / format	Meaning
Export origin	Objects should have been exported using database explorer export before import.
Privileges	Destination user needs appropriate INSERT/UPDATE, DROP, CREATE object privileges and the IMPORT system privilege.
Formats	CSV and PARQUET are supported import formats.
Source locations	Local computer, Amazon S3, Azure Storage, Alibaba Cloud OSS, Google Cloud Storage, and Data Lake Files.
Compatibility caution	On-premise binary or CSV exports may not be compatible with SAP HANA Cloud import due to format and data-type differences.

Cockpit opening path: From Cloud Central, select the instance actions menu (...) and choose Open in SAP HANA Cockpit. The cockpit overview opens in a new browser tab and shows monitoring cards.

Module 7 - Lifecycle Operations: Start, Stop, Upgrade, Backup, Recovery

Objective: Perform basic lifecycle operations on SAP HANA Cloud database instances and understand when to use Cloud Central versus CLI commands.

SAP HANA Cloud Instance Lifecycle Operations



Cloud Central actions and Cloud Foundry CLI can both control start/stop; upgrades involve a restart window.

Figure 8. Instance lifecycle operations.

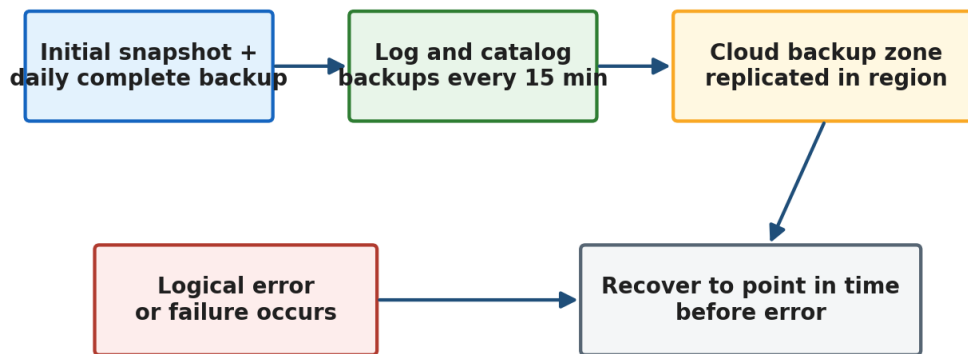
Operation	Cloud Central path	Important note
Stop	Cloud Central > instance actions (...) > Stop.	Stops the database instance; training text describes shutdown as roughly 30 seconds.
Start	Cloud Central > instance actions (...) > Start.	Starts the database instance; startup can take several minutes.
Upgrade / Patch	Cloud Central > instance actions (...) > Upgrade or Patch.	Action is active only when a new version or revision is available; operation involves a restart unless patch configuration allows otherwise.
Check version	Open SAP HANA cockpit and search for Database Information.	Review version, version history, and installed plugin versions before and after upgrade.

```
# Stop a SAP HANA Cloud database service with Cloud Foundry CLI
cf update-service <service name> -c '{"data": {"serviceStopped": true}}'

# Start a SAP HANA Cloud database service with Cloud Foundry CLI
cf update-service <service name> -c '{"data": {"serviceStopped": false}}'

# Example
cf update-service HC200-Demo-Database -c '{"data": {"serviceStopped": false}}'
```

Backup and Point-in-Time Recovery Model



Best practice: investigate root cause before recovery, then recover to the safest point in time.

Figure 9. Backup and point-in-time recovery model.

Backup component	Study note
Initial backup	Initial snapshot and complete data backup are created on instance creation.
Complete data backup	Performed automatically every 24 hours while the database is online.
Log backups	Performed automatically every 15 minutes.
Catalog backups	Performed automatically every 15 minutes.
Retention	Default retention is 14 daily backups / generations, but can be changed depending on configuration.
RPO	Training text states an RPO of no more than 15 minutes.

Recovery best practice: For logical errors, recover to a point in time immediately before the error occurred. First investigate root cause; recovery fixes state, not necessarily the cause.

Module 8 - Resilience, Scaling, Workload, and Resources

Objective: Explain how regions, availability zones, replicas, scaling, resource monitoring, and workload monitoring support reliable SAP HANA Cloud operations.

Concept	Definition / value
Region	Geographical location where applications, data, or services are hosted. Each subaccount has exactly one region and the assignment cannot be changed.
Availability zone	Independent failure domain in a region. Multiple zones in a region improve fault tolerance.
Replica	Exact copy of a source SAP HANA Cloud database. One or two replicas can reduce outage due to maintenance, faults, or disasters in productive landscapes.
Synchronous same-zone replica	Best takeover time and high availability; no disaster tolerance; RPO can be zero.
Different-zone replica	Improves disaster tolerance, but network latency may increase.

Account limitation: Replica features are not available in free tier or trial SAP BTP accounts. They are supported in productive landscapes and add cost.

Scaling action	System behavior
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Scale up memory	Increases memory and also increases vCPU and disk values according to ratios.
Scale down memory	Reduces vCPUs, but disk storage remains unchanged to avoid data loss.
Scale storage up/down	Changes storage without changing memory or vCPU count.
Trial/free tier	Self-service scaling is not available because size is fixed.

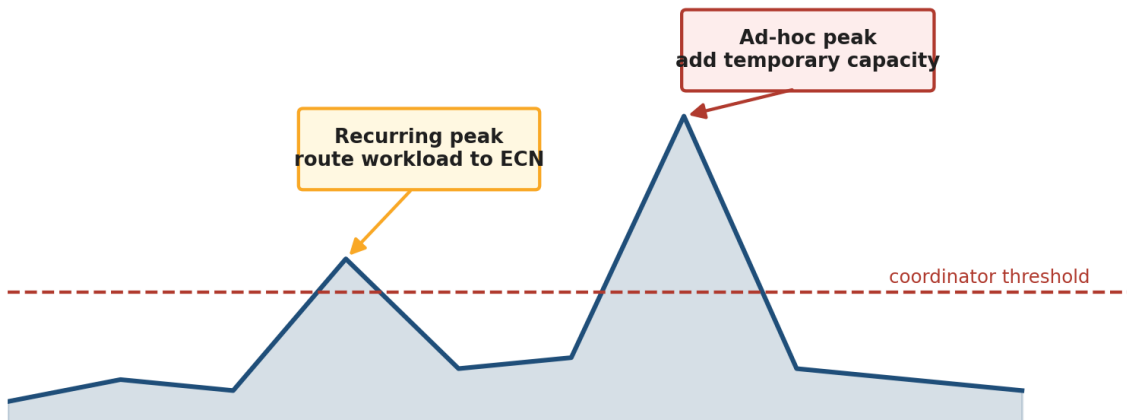
Monitoring card / app	What it helps you analyze
Services	Service status, CPU, memory, ports, process IDs, and service-level health.
Performance Monitor	Historical memory, CPU, storage, network, transactions, and threads over selected time windows.
Sessions	Connected applications/users, active or idle sessions, blocking, query runtime, DML/DDDL counts.
Threads	Long-running or blocked threads and current execution activity.
SQL Statements	Expensive, frequent, or problematic SQL workload patterns.

Monitoring method: Use KPI cards as entry points. A mini-chart is not the final analysis; click through to the full monitor to inspect timeframe, host, and metric detail.

Module 9 - Elastic Compute Nodes

Objective: Explain why ECNs exist, how to provision and monitor them, how workload classes route queries, and how ECN Advisor recommendations are used.

Elastic Compute Nodes: Cost-Efficient Peak Handling



ECNs add temporary compute capacity; use workload classes and optional table replication to route workload.

Figure 10. Elastic Compute Nodes for temporary peak workload handling.

ECN topic	Key point
Why ECNs exist	Sizing a database for rare peak workload is expensive. ECNs provide temporary compute capacity during recurring or ad-hoc peaks.
What an ECN is	Additional instance running a computeserver process with temporary persistence.
When cost applies	Additional cost is incurred while the computeserver / ECN is running.
Functional areas	Provisioning, workload classes, query routing, table replication, table placement, and redistribution.
Admin support	Cloud Central includes ECN sizing, cost estimation, workload analysis, recommendations, and monitoring tools.

Setup step	Required idea
Estimate resources	Use the SAP HANA Cloud Capacity Unit Estimator to estimate CU requirements based on compute, memory, storage, and hours.
Add ECN	Instance Overview > Manage Configuration > Additional Nodes > Add ECN.
Provision ECN	Give name, performance class, memory, and storage. Use "ecn" in the name for easy identification.
Route workload	Create or edit workload classes and set Routing Location to the ECN.
Optional replication	Replicate tables or define table placement/redistribution when query performance requires local data.

-- Route workload class to an ECN location and allow result lag for async replica usage

```
CREATE WORKLOAD CLASS "MyWLC"
SET 'ROUTING LOCATION HINT' = 'ecn_group1',
    'RESULT LAG HINT' = 'HANA_ATR';
```

-- Map all queries from a named user to the workload class

```
CREATE WORKLOAD MAPPING "MyWLCMapping1"
WORKLOAD CLASS "MyWLC"
SET 'USER NAME' = 'ECN_USER';
```

-- Map by application component name

```
CREATE WORKLOAD MAPPING "MyWLCMapping2"
WORKLOAD CLASS "MyWLC"
SET 'APPLICATION COMPONENT NAME' = 'My_APP';
```

Provisioning sequence: Provision ECN first, then add routing location to the workload class. For deprovisioning, remove routing from the workload class first, then deprovision the ECN.

ECN Advisor point	Details
Purpose	Analyzes coordinator workload and recommends ECN usage to bring workload below a threshold.
Default state	Switched off by default.
Prerequisites	Minimum five vCPUs, at least one workload class, viewer role to access data, administrator role to configure and request recommendations.
Timeframe	Choose a 30-minute to 24-hour timeframe that ends at least 20 minutes before recommendation request and falls within the previous 14 days.
Recommendation validity	Recommendations remain valid for 100 days after creation.

Module 10 - Security: Auditing, Users, User Groups, and Roles

Objective: Set up core governance by understanding audit policies, audit trails, database users, user groups, and role-based authorization.

Security and Governance Building Blocks

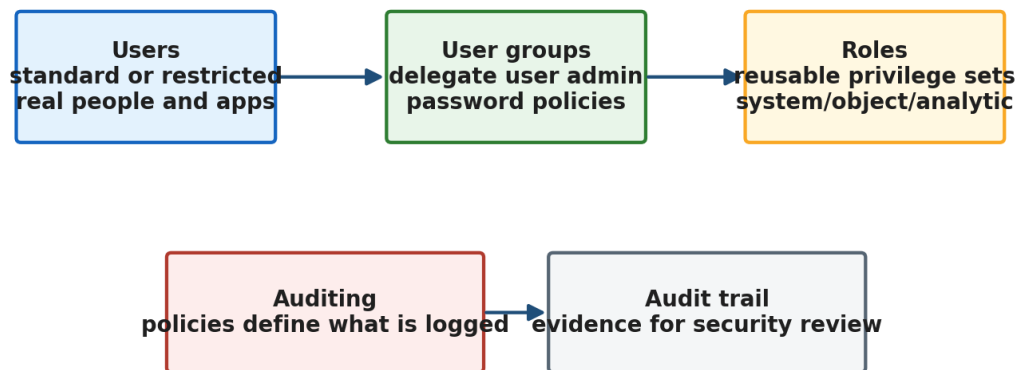


Figure 11. Security and governance building blocks.

Auditing topic	Key study note
Purpose	Auditing records important database actions such as sensitive data access, object changes, authentication, authorization changes, and system configuration changes.
Default state	Audit logging is disabled by default.
Configuration tools	Use SAP HANA cockpit or SQL to enable auditing, configure trail targets, and create policies.
Mandatory audit actions	When auditing is active, changes to audit policies, enabling/disabling policies, and deletion of audit entries are always audited.
Audit trail access	Audit entries are accessible through the AUDIT_LOG public system view if the user has AUDIT OPERATOR, AUDIT READ, or AUDIT ADMIN.

-- Create and activate a simple audit policy for connection events
 CREATE AUDIT POLICY "HC200 Demo Audit Policy" AUDITING ALL CONNECT LEVEL INFO;
 ALTER AUDIT POLICY "HC200 Demo Audit Policy" ENABLE;

Important limitation: Only actions inside the database engine can be audited. If the database is offline when an action occurs, that action cannot be detected by the database auditing mechanism.

User concept	Definition / usage
Standard users	Created with CREATE USER. Can create objects in their own schema and read system views through the PUBLIC role by default.
Restricted users	Created with CREATE RESTRICTED USER. Initially have no privileges and are intended for users accessing SAP HANA through client applications.
User groups	Manage related users together, delegate user management, and apply group-specific configurations such as password policy or client connect restrictions.
Group administrators	Designated administrators manage users in a group; useful for separating technical, privileged, or department-specific users.
Data access	User groups do not control data access. Roles and privileges control data access.

-- User group administration examples
 GRANT USERGROUP OPERATOR ON USERGROUP TechnicalUsers
 TO TechnicalUsersAdmin WITH GRANT OPTION;

```
CREATE USER <username> SET USERGROUP <usergroupname>;
ALTER USER <username> SET USERGROUP <usergroupname>;
```

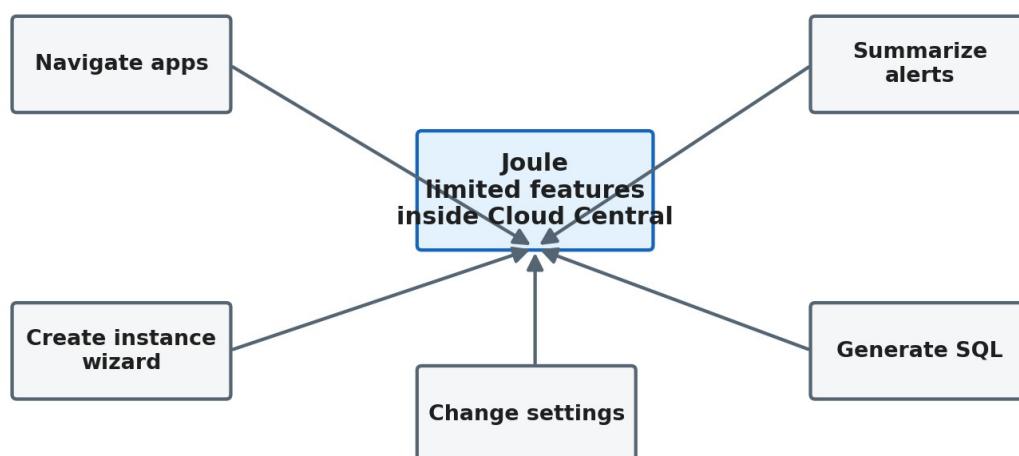
Privilege / role type	Meaning
System privileges	Control general system activities such as creating schemas, users, roles, backups, and licenses.
Object privileges	Control access to database objects such as SELECT, CREATE ANY, ALTER, DROP, INSERT, UPDATE, DELETE.
Analytic privileges	Control read access to data in information models based on values or combinations of values.
Catalog roles	Runtime roles created and assigned with SQL or SAP HANA cockpit.
Design-time roles	Roles modeled as deployable artifacts, for example .hdbrole files.

Best practice: Grant privileges through roles rather than directly to users. Roles make authorization reusable, easier to audit, and aligned to business tasks.

Module 11 - Gen AI and Joule in SAP HANA Cloud Central

Objective: Understand how to activate and use Joule with limited features in SAP HANA Cloud Central, including its scope and limitations.

Joule in SAP HANA Cloud Central



Treat AI output as assistance: verify SQL, permissions, and operational impact before execution.

Figure 12. Joule use cases and verification reminder.

Joule capability	Example prompt / use
Create instance flow	"Create a database instance" opens or assists with the provisioning wizard.
Navigate apps	"Open SQL Console", "Open database migration", "Open database objects", or "Open Alerts".
Summarize alerts	"Show Alerts Overview" creates an alert summary and supports troubleshooting.
Generate SQL from plain text	"Display all audit policies using SQL" or SQL generation prompts for schemas, tables, inserts, and selects.
Find documentation	Ask about SAP HANA Cloud features; Joule can provide explanations and documentation links.
Change user settings	Prompts such as changing the theme, preferred region, or app notifications can return action links.

Activation step	Detail
Role check	Verify SAP HANA Cloud Administrator role collection is assigned in the subaccount.
Open Cloud Central	Global Account > Subaccount > Instances and Subscriptions > SAP HANA Cloud.
Open user settings	Click the user icon at the top-right and select Settings.
Enable Gen AI	Select Gen AI, set the toggle to Enabled, read and accept terms, then save.
Scope	Enabling Gen AI applies to all users within the subaccount.

Limitations: Joule with limited features in SAP HANA Cloud Central is isolated to the SAP HANA Cloud context. It does not provide the full Joule AI copilot capabilities, broad SAP integration, role-based access features, or digital assistant functionality.

-- Prompt examples for Joule
 Change the theme to "SAP morning Horizon"
 Change the Preferred Region to "Americas"
 Change the App Notifications to "On"
 Open SQL Console
 Open database objects
 Show Alerts Overview
 Show the SQL to create a schema called HC200_SCHEMA

Verification habit: Treat Joule output as a productivity assistant. Always verify SQL syntax, target schema, permissions, data impact, and operational risk before execution.

Exercise and Demo Checklist

Use this checklist to track hands-on practice. Mark each item after you complete it in the training system.

Area	Activity	Type
Account access	Log in to enterprise or trial cockpit and identify the global account and subaccount.	Demo / practice
Entitlements	Verify hana, relational-data-lake, tools, and Cloud Foundry runtime entitlement needs.	Practice
Open SAP HANA Cloud Central	BTP Cockpit > Subaccount > Instances and Subscriptions > SAP HANA Cloud.	Demo
Display database details	Open instance overview and inspect Memory, Compute, and Consumption cards.	Demo
Add integrated data lake	Use instance actions > Add Data Lake; configure name, HDLADMIN password, size, connections, and backups.	Demo + Exercise
Add replicas	Use Cloud Central to add one synchronous and one asynchronous replica where supported.	Exercise
Check database alerts	Open Alerts app, filter Error/Warning alerts, inspect details and recommended action.	Demo / Exercise
Configure alert rules	Open Configure Alert Rules, select instance, edit thresholds, interval, and enabled status.	Exercise
Implement alert rule set	Create alert rule set, add rules, add instances, review, and Save or Save and Apply.	Exercise
Configure alert notifications	Use SAP BTP Alert Notification Service to route events to email, Teams, Slack, ServiceNow, or webhook.	Tutorial / Exercise
Monitor performance	Open Performance Monitor, select instance/host/timeframe, and add custom KPI chart.	Demo
Use SQL Console	Connect to instance, choose schema, run SQL, inspect results, history, and background statements.	Demo / Exercise
Use SQL Statement Library	Format SQL, save statement with name/description, reopen it from library, execute it.	Demo / Exercise
Open Database Explorer	Cloud Central > instance actions > Open in SAP HANA Database Explorer.	Exercise
Examine traces	Browse Database Diagnostic Files > indexserver, open trace, search for warnings/errors.	Exercise
Create full system dump	Open Manage full system information dumps, collect and download ZIP for support.	Demo / Practice
Import catalog objects	Verify privileges, choose import source, import CSV/PARQUET/catalog package.	Exercise
Open HANA Cockpit	Cloud Central > instance actions > Open in SAP HANA Cockpit.	Exercise
Stop database	Use Cloud Central Stop or cf update-service with serviceStopped true.	Exercise
Start database	Use Cloud Central Start or cf update-service with serviceStopped false.	Exercise
Upgrade / patch database	Check version, run Upgrade/Patch, verify version history after restart.	Exercise
Verify backups	Open Database Backups card in cockpit and review backup catalog/generations.	Practice
Recover database	Perform root cause check, choose point in time, run recovery process.	Exercise
Set up auditing	Run basic setup or create audit policy with SQL and inspect AUDIT_LOG.	Exercise

Create user groups	Create group, assign group admins, configure password policy or connect restriction.	Practice
Create roles	Create catalog role and assign system/object/analytic privileges to users.	Practice
Set up ECN	Estimate capacity, add ECN, create workload class, route workload, monitor ECN.	Demo / Exercise
Use ECN Advisor	Activate advisor, select timeframe, get recommendation, apply routing and ECN changes.	Demo / Exercise
Activate Gen AI	Settings > Gen AI > Enabled, accept terms, save.	Demo
Use Joule	Open tools, summarize alerts, change settings, generate SQL, verify outputs.	Demo / Exercise

Practice Questions

1. Which account type should an individual student use to learn SAP HANA Cloud?

- Answer: SAP BTP trial account.

2. What is the preferred tool for administering SAP HANA Cloud instances?

- Answer: SAP HANA Cloud Central.

3. Which tool is best for browsing catalog objects and diagnostic files?

- Answer: SAP HANA database explorer.

4. What is the difference between an alert rule and an alert rule set?

- Answer: An alert rule defines one monitoring check and threshold behavior; an alert rule set groups multiple rules for reuse and application to instances.

5. Why should you avoid allowing all IP addresses for data lake connections?

- Answer: It exposes the instance to the internet and increases security risk.

6. What happens when memory is scaled down?

- Answer: vCPUs are reduced, but disk storage is not reduced to avoid data loss.

7. What is a backup generation?

- Answer: The complete set of backup types created from one complete data backup until the next complete data backup.

8. Why investigate before database recovery?

- Answer: Recovery restores state but may not fix the root cause of the incident.

9. What problem do ECNs solve?

- Answer: They add temporary compute capacity for peak workloads without permanently sizing the database for rare peaks.

10. What controls data access: user groups or roles/privileges?

- Answer: Roles and privileges control data access; user groups organize user administration.

11. What SQL view can be used to read audit trail entries?

- Answer: AUDIT_LOG, with required audit privileges.

12. What is the biggest rule when using Joule-generated SQL?

- Answer: Verify SQL, permissions, schema, and data impact before executing.

Command and SQL Cheat Sheets

```
# SAP BTP Cockpit direct access examples
EMEA: https://emea.cockpit.btp.cloud.sap
EU Access: https://eu-access.cockpit.btp.cloud.sap
Americas: https://amer.cockpit.btp.cloud.sap
APAC: https://apac.cockpit.btp.cloud.sap
Trial: https://account.hanatrial.ondemand.com/trial/#/home/trial
```

```
# Cloud Foundry CLI
cf login -o <organisation> -s <space> -a <API endpoint>
cf update-service <service name> -c '{"data":{"serviceStopped": true}}'
cf update-service <service name> -c '{"data":{"serviceStopped": false}}'
```

```
-- Auditing
CREATE AUDIT POLICY "HC200 Demo Audit Policy" AUDITING ALL CONNECT LEVEL INFO;
ALTER AUDIT POLICY "HC200 Demo Audit Policy" ENABLE;
SELECT * FROM AUDIT_LOG;

-- SQL basics
CREATE SCHEMA "HC200_SCHEMA";
SET SCHEMA "HC200_SCHEMA";
CREATE COLUMN TABLE "HC200_SCHEMA"."HC200_DEMO" (
  "ProductNumber" INT PRIMARY KEY,
  "ProductName" NVARCHAR(255),
  "Stock" INT
);
INSERT INTO "HC200_DEMO" VALUES (1, 'Bike', 15);
SELECT "ProductName", "Stock" FROM "HC200_DEMO" WHERE "Stock" < 5;

-- ECN workload routing
CREATE WORKLOAD CLASS "MyWLC" SET 'ROUTING LOCATION HINT' = 'ecn_group1';
CREATE WORKLOAD MAPPING "MyWLCMapping1" WORKLOAD CLASS "MyWLC" SET 'USER NAME' = 'ECN_USER';
```

Selected Further Reading Links Mentioned in the Lesson Text

- [SAP Help Portal - Enterprise accounts](#)
- [SAP Developer Tutorial - Create trial account](#)
- [SAP Developer Tutorial - SAP HANA Cloud alerts](#)
- [Cloud Foundry CLI downloads](#)
- [Kubernetes documentation](#)

Source Note

These notes were synthesized from the user-supplied SAP HANA Cloud lesson text for Sprint 2 and the uploaded supporting text files. The diagrams are original study aids created from the supplied material.